$$|d\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \qquad (2.12)$$

With these column vectors in hand, it will be easy to create column vectors for |*r*> and |*l*> using Eqs. 2.5 and 2.6, and for |*i*> and |*o*> using Eqs. 2.10. We'll do that in the next lecture, where these results are needed.

2.7 PUTTING IT ALL TOGETHER

We have covered a lot of ground in this lecture. Before moving on, let's take stock of what we've done. Our goal was to synthesize what we know about spins and vector spaces. We figured out how to use vectors to represent spin states, and in the process we got a glimpse of the kind of information a state-vector contains (and does *not* contain!). Here is a brief outline of what we did:

- Based on our knowledge of spin measurements, we chose three pairs of mutually orthogonal basis vectors. Pair-wise, we named them |*u*> and |*d*>, |*r*> and |*l*>, and |*i*> and |*o*>. Because the basis vectors |*u*> and |*d*> represent physically distinct states, we were able to assert that they are mutually orthogonal. In other words, *u*|*d*> = 0. The same holds for |*r*> and |*l*>, and also for |*i*> and |*o*>.

- We found that it takes two independent parameters to specify a spin state, and then we arbitrarily chose one of the orthogonal pairs, |*u*> and |*d*>, as our basis vectors for representing *all* spin states—even though the two complex numbers in a state-vector require four real numbers to specify them. How did we get away with this? We were clever enough to notice that these four numbers are not all independent.[4] The normalization constraint (total probability must equal 1) eliminates one independent parameter, and "phase indifference" (the physics of a state-vector